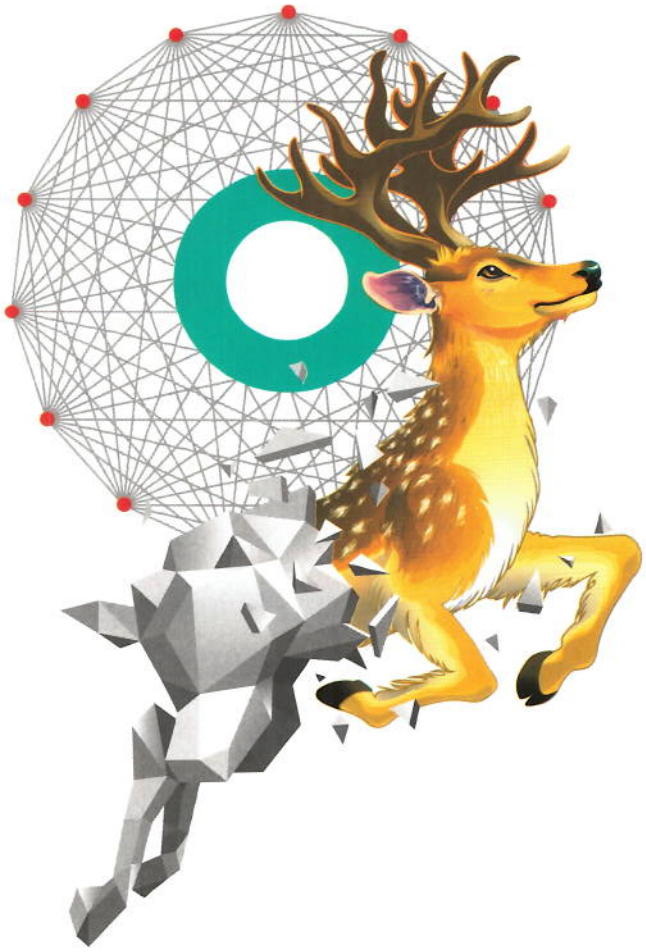
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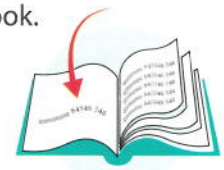
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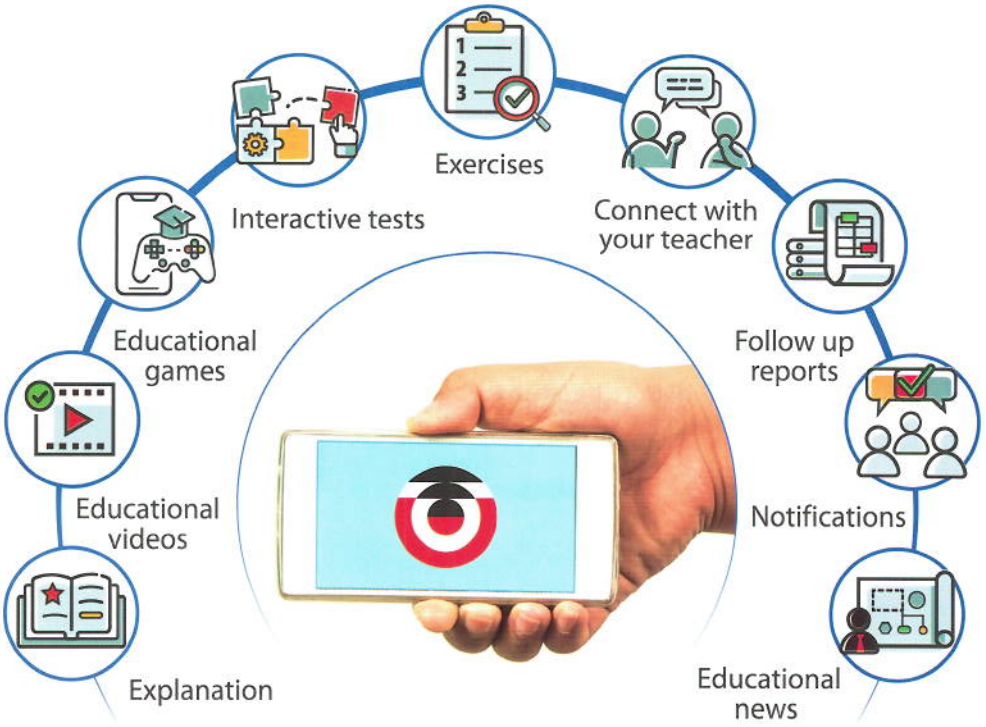
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CONTENTS

First | Algebra and Analytic Solid Geometry

Unit 1 Binomial theorem

Unit 2 Complex numbers

Unit 3 Geometry and measurement in three dimensions

Unit 4 Straight lines and planes in space

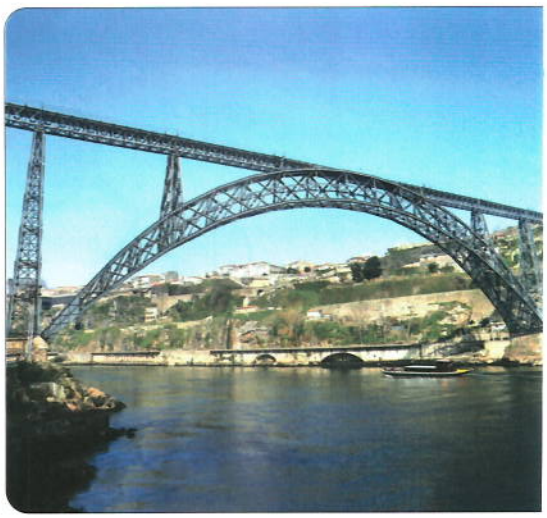


Second | Differential and Integral Calculus

Unit 1 Differentiation and its applications.

Unit 2 Behavior of the function and curve sketching.

Unit 3 The definite integral and its applications.



First

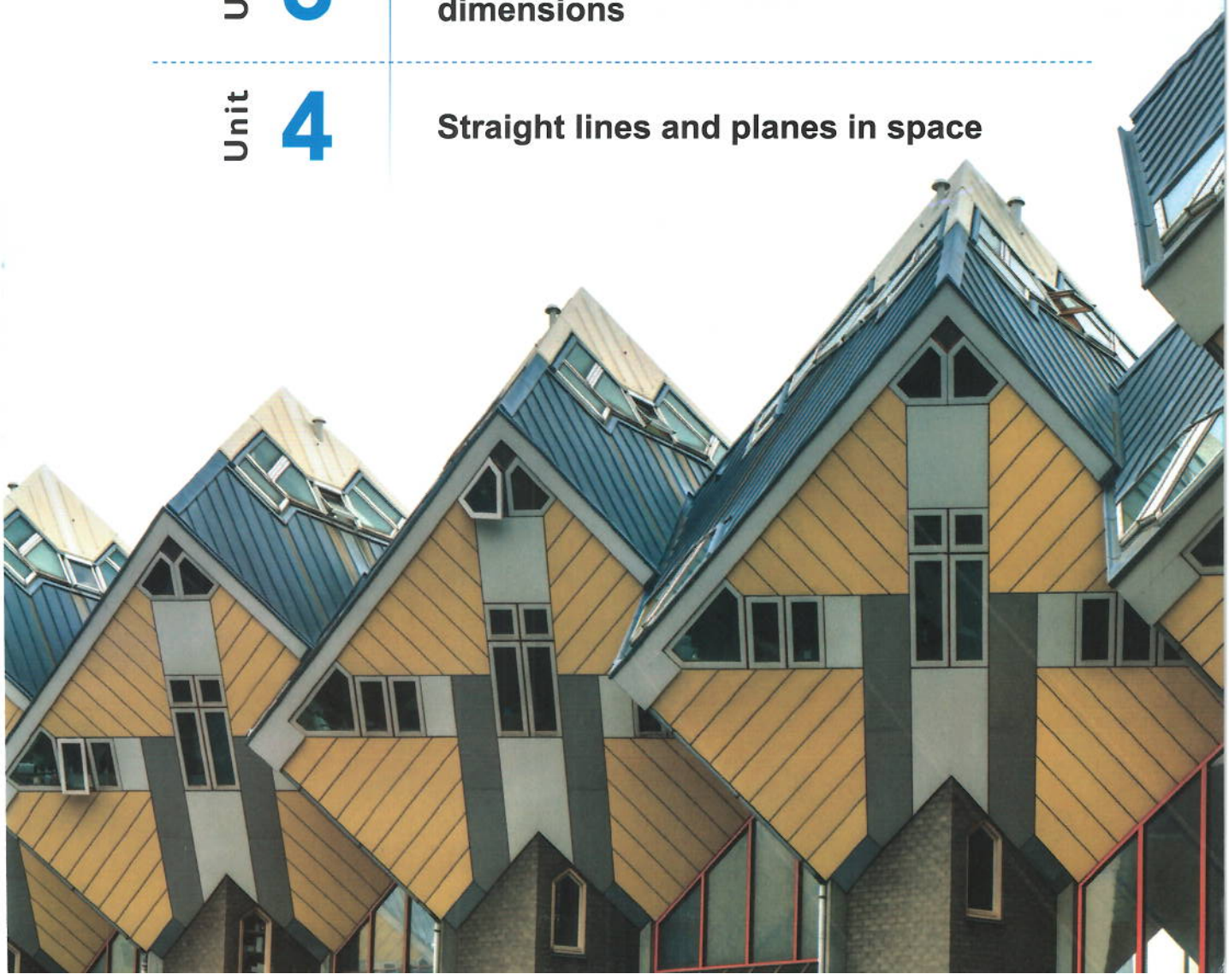
Algebra and Analytic Solid Geometry

Unit 1 Binomial theorem

Unit 2 Complex numbers

Unit 3 Geometry and measurement in three dimensions

Unit 4 Straight lines and planes in space



UNIT

ONE

Binomial theorem

Lesson 1 Binomial theorem for positive integer power.

Lesson 2 Finding the term containing x^r in the expansion of binomial.

Lesson 3 Ratio between two consecutive terms in the binomial expansion.



Unit 1

Lesson One

Binomial theorem for positive integer power

Binomial theorem

If x , $a \in \mathbb{R}$, n is any positive integer , then :

1. $(x+a)^n = {}^nC_0\, x^n + {}^nC_1\, x^{n-1}\, a + {}^nC_2\, x^{n-2}\, a^2 + \ldots\ldots\ldots + {}^nC_r\, x^{n-r}\, a^r + \ldots\ldots\ldots + {}^nC_n\, a^n$

and the right hand side is called the expansion of the left hand side and the expansion is arranged in a descending order according to the exponents of x and arranged in an ascending order according to the exponents of a

2. $(x-a)^n = (x+(-a))^n$

$= {}^nC_0\, x^n + {}^nC_1\, x^{n-1}\, (-a) + {}^nC_2\, x^{n-2}\, (-a)^2 + \ldots\ldots\ldots + {}^nC_n\, (-a)^n$

$= {}^nC_0\, x^n - {}^nC_1\, x^{n-1}\, a + {}^nC_2\, x^{n-2}\, a^2 + \ldots\ldots\ldots + {}^nC_n\, (-a)^n$

Notice that : The sign of any term in the expansion differs from the sign of the previous term and the sign of the last term will be positive if n is an even number and negative if n is an odd number.

For example :

* $(x+a)^2 = {}^2C_0\, x^2\, a^0 + {}^2C_1\, x^{2-1}\, a^1 + {}^2C_2\, x^{2-2}\, a^2$

$= x^2 + 2\,x\,a + a^2$

* $(x-a)^3 = {}^3C_0\, x^3\, a^0 - {}^3C_1\, x^{3-1}\, a^1 + {}^3C_2\, x^{3-2}\, a^2 - {}^3C_3\, x^{3-3}\, a^3$

$= x^3 - 3\,x^2\,a + 3\,x\,a^2 - a^3$

* $(x \pm a)^4 = x^4 \pm 4\,x^3\,a + 6\,x^2\,a^2 \pm 4\,x\,a^3 + a^4$ and so on

Remarks

1. In the expansion of $(x \pm a)^n$

- The number of terms of the expansion is more than the power of the algebraic expression by one

i.e. The number of terms of the expansion = (n + 1) terms

- The exponents of "x" decreases gradually by one , the exponents of "a" increases gradually by one , such that the sum of the two exponents of x and a in any term equals n

i.e. The power of x + the power of a = n (in any term)

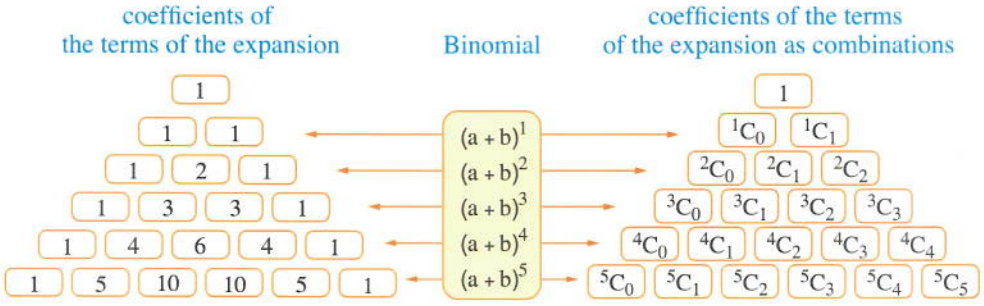
- The index (value of r) in ${}^{n}C_{r}$ of each term is less than the order of the term by one.

For example : $T_{\text{⑤the order of the term}} = {}^{n}C_{\text{④the index}}\ x^{n-4}\ a^{4}$

The first term : $T_{1} = {}^{n}C_{0}\ x^{n}\ a^{0} = x^{n}$

The last term : $T_{n+1} = {}^{n}C_{n}\ x^{0}\ (\pm a)^{n} = (\pm a)^{n}$

2. The coefficients of the expansion of the expression $(a + b)^{n}$ follows a pattern in Pascal's triangle.



Example 1

Write the expansion of : $(2x - 3y)^{5}$

Solution

$$(2x - 3y)^{5} = {}^{5}C_{0}\ (2x)^{5}\ (3y)^{0} - {}^{5}C_{1}\ (2x)^{4}\ (3y)^{1} + {}^{5}C_{2}\ (2x)^{3}\ (3y)^{2}$$

$$- {}^{5}C_{3}\ (2x)^{2}\ (3y)^{3} + {}^{5}C_{4}\ (2x)^{1}\ (3y)^{4} - {}^{5}C_{5}\ (2x)^{0}\ (3y)^{5}$$

$$= 32x^{5} - 5\ (16x^{4})\ (3y) + 10\ (8x^{3})\ (9y^{2}) - 10\ (4x^{2})\ (27y^{3})$$

$$+ 5\ (2x)\ (81y^{4}) - 243y^{5}$$

$$= 32x^{5} - 240x^{4}y + 720x^{3}y^{2} - 1080x^{2}y^{3} + 810xy^{4} - 243y^{5}$$

Remarks

- If we want to find the sum of the terms coefficient of expansion in a binomial expression , we have to put every value of variable in the binomial equals whole one without finding the expansion.

For example : (In the previous example)

1. The sum of the terms coefficient of expansion $(2x - 3y)^5$

$= 32 - 240 + 720 - 1080 + 810 - 243 = -1$

and we can find the sum of the coefficients directly by putting $x = 1$, $y = 1$ in the binomial $(2x - 3y)^5$

i.e. : The sum of the terms coefficients of the expansion of $(2x - 3y)^5 = (2 - 3)^5 = -1$

Special cases of the expansion of a binomial :

If $x \in \mathbb{R}$, n is positive integer , then :

1. $(1 + x)^n = 1 + {}^nC_1\, x + {}^nC_2\, x^2 + {}^nC_3\, x^3 + \ldots\ldots\ldots + x^n$

2. $(1 - x)^n = 1 - {}^nC_1\, x + {}^nC_2\, x^2 - {}^nC_3\, x^3 + \ldots\ldots\ldots + (-x)^n$

For example :

- $(1 \pm x)^5 = 1 \pm {}^5C_1\, x + {}^5C_2\, x^2 \pm {}^5C_3\, x^3 + {}^5C_4\, x^4 \pm {}^5C_5\, x^5$

$= 1 \pm 5x + 10x^2 \pm 10x^3 + 5x^4 \pm x^5$

Example 2

Write the expansion of $(1 + x)^7$, then use it to :

1. Prove that : ${}^7C_0 + {}^7C_1 + {}^7C_2 + {}^7C_3 + \ldots\ldots\ldots + {}^7C_7 = 2^7$

2. Prove that : ${}^7C_0 - {}^7C_1 + {}^7C_2 - {}^7C_3 + \ldots\ldots\ldots - {}^7C_7 = 0$

3. Find the value of : $1 + 3 \times {}^7C_1 + 9 \times {}^7C_2 + 27 \times {}^7C_3 + \ldots\ldots\ldots + 3^7$

Solution

$(1 + x)^7 = {}^7C_0 + {}^7C_1\, x + {}^7C_2\, x^2 + {}^7C_3\, x^3 + \ldots\ldots\ldots + {}^7C_7\, x^7$

1. Put $x = 1$

$\therefore (1 + 1)^7 = {}^7C_0 + {}^7C_1 + {}^7C_2 + {}^7C_3 + \ldots\ldots\ldots + {}^7C_7$

$\therefore {}^7C_0 + {}^7C_1 + {}^7C_2 + {}^7C_3 + \ldots\ldots\ldots + {}^7C_7 = 2^7$

2. Put $x = -1$

$\therefore (1-1)^7 = {}^7C_0 - {}^7C_1 + {}^7C_2 - {}^7C_3 + \ldots\ldots\ldots - {}^7C_7$

$\therefore {}^7C_0 - {}^7C_1 + {}^7C_2 - {}^7C_3 + \ldots\ldots\ldots - {}^7C_7 = 0$

3. Put $x = 3$

$\therefore (1+3)^7 = {}^7C_0 + {}^7C_1 \times 3 + {}^7C_2 \times 9 + {}^7C_3 \times 27 + \ldots\ldots\ldots + {}^7C_7 \times 3^7$

$\therefore 1 + 3 \times {}^7C_1 + 9 \times {}^7C_2 + 27 \times {}^7C_3 + \ldots\ldots\ldots + 3^7 = 4^7$

Example 3

Use the expansion of : $(1+x)^n$ in proving that :

${}^nC_1 + 2 \times {}^nC_2 + 3 \times {}^nC_3 + 4 \times {}^nC_4 + \ldots\ldots\ldots + n \times {}^nC_n = n \times 2^{n-1}$

Solution

$\because (1+x)^n = 1 + {}^nC_1\, x + {}^nC_2\, x^2 + {}^nC_3\, x^3 + {}^nC_4\, x^4 + \ldots\ldots\ldots + {}^nC_n\, x^n$

By differentiating the two sides by x :

$\therefore n(1+x)^{n-1} = 0 + {}^nC_1 + 2\,{}^nC_2\, x + 3\,{}^nC_3\, x^2 + 4\,{}^nC_4\, x^3 + \ldots\ldots\ldots + n\,{}^nC_n\, x^{n-1}$

Put $x = 1$

$\therefore n(1+1)^{n-1} = {}^nC_1 + 2\,{}^nC_2 + 3\,{}^nC_3 + 4\,{}^nC_4 + \ldots\ldots\ldots + n\,{}^nC_n$

$\therefore {}^nC_1 + 2\,{}^nC_2 + 3\,{}^nC_3 + 4\,{}^nC_4 + \ldots\ldots\ldots + n\,{}^nC_n = n \times 2^{n-1}$

Another solution :

By finding the expansion of : $(1+x)^{n-1}$, then

puting $x = 1$

$\therefore 2^{n-1} = {}^{n-1}C_0 + {}^{n-1}C_1 + {}^{n-1}C_2 + \ldots\ldots\ldots + {}^{n-1}C_{n-1}$

, $\because {}^nC_1 + 2 \times {}^nC_2 + 3 \times {}^nC_3 + \ldots\ldots\ldots + n \times {}^nC_n$

$= \sum_{r=1}^{n} r \times {}^nC_r$

$= \sum_{r=1}^{n} \cancel{r} \times \frac{n}{\cancel{r}} \times {}^{n-1}C_{r-1}$

$= n \sum_{r=1}^{n} {}^{n-1}C_{r-1}$

$= n({}^{n-1}C_0 + {}^{n-1}C_1 + {}^{n-1}C_2 + \ldots\ldots\ldots + {}^{n-1}C_{n-1}) = n \times 2^{n-1}$

Notice that

$${}^nC_r = \frac{\lfloor n}{\lfloor r \; \lfloor n-r} = \frac{n \; \lfloor n-1}{r \; \lfloor r-1 \; \lfloor n-r}$$

$$= \frac{n}{r} \times {}^{n-1}C_{r-1}$$

Example 4

Write the expansion of each of the following :

$(x-1)^5\,(x+1)^5$

Solution

$$(x-1)^5\,(x+1)^5 = ((x-1)\,(x+1))^5 = (x^2-1)^5$$

$$= (x^2)^5 - {}^5C_1\,(x^2)^4 + {}^5C_2\,(x^2)^3 - {}^5C_3\,(x^2)^2 + {}^5C_4\,(x^2) - {}^5C_5$$

$$= x^{10} - 5x^8 + 10x^6 - 10x^4 + 5x^2 - 1$$

The general term in the expansion of a binomial

The general term in the expansion of : $(x+a)^n$

is denoted by T_{r+1} where : $r \in \mathbb{N}$, $r \leq n$

hence : $T_{r+1} = {}^nC_r\,a^r\,x^{n-r}$

i.e. The general term of the binomial expansion $= {}^nC_r \times (\text{second})^r \times (\text{first})^{n-r}$

Example 5

Find the sixth term according to the descending exponents of x in the expansion of :

$$\left(3x^2 - \frac{1}{3x}\right)^9$$

Solution

$\because T_{r+1} = {}^nC_r \times (2^{nd})^r \times (1^{st})^{n-r}$

$$\therefore T_6 = {}^9C_5 \times \left(\frac{-1}{3x}\right)^5 \times (3x^2)^4 = 126 \times \frac{-1}{243\,x^5} \times 81\,x^8 = -42\,x^3$$

Example 6

If the ninth term in the expansion of : $(x^2 - x^{-2})^{11}$ according to descending exponents of x equals 165 , find the value of : x

Solution

$\because T_9 = {}^{11}C_8\,(-x^{-2})^8\,(x^2)^3 = 165\,x^{-16} \times x^6 = 165\,x^{-10}$ $\quad\therefore 165\,x^{-10} = 165$

$\therefore x^{-10} = 1$ $\quad\therefore x^{10} = 1$ $\quad\therefore x = \pm 1$

Example 7

In the expansion of : $\left(3x^2 - \frac{1}{x}\right)^{13}$, find the tenth term from the end. according to the descending exponents of x

Solution

The tenth term from the end in the expansion of : $\left(3x^2 - \frac{1}{x}\right)^{13}$ is the tenth term from the beginning in the expansion of : $\left(\frac{-1}{x} + 3x^2\right)^{13}$

, $\because T_{10}$ in the expansion of : $\left(\frac{-1}{x} + 3x^2\right)^{13} = {}^{13}C_9 (3x^2)^9 \left(\frac{-1}{x}\right)^4$

$$= 715 \times 3^9 \times x^{18} \times \frac{1}{x^4} = 715 \times 3^9 x^{14}$$

$\therefore$ The tenth term from the end in the expansion of : $\left(3x^2 - \frac{1}{x}\right)^{13} = 715 \times 3^9 x^{14}$

Remark

If the order of the term is known from the end in the expansion of a binomial , then : the order of this term from the beginning = number of terms of the expansion – The order of the term from the end + 1

- **Another solution of the previous example :**

$\because$ The order of the term from the end is 10th

$\therefore$ The order of this term from the beginning

= 14 (number of terms of the expansion) – 10 (the order of the term from the end) + 1 = 5

i.e. The 10th term from the end is T_5 from the beginning in the expansion of $\left(3x^2 - \frac{1}{x}\right)^{13}$

, $\because T_5 = {}^{13}C_4 \times \left(\frac{-1}{x}\right)^4 \times (3x^2)^9 = 715 \times \frac{1}{x^4} \times 3^9 x^{18} = 715 \times 3^9 x^{14}$

$\therefore$ The 10th term from the end = $715 \times 3^9 x^{14}$

Example 8

Find the 5th term in the expansion of :

$(2 + x)^8 - {}^8C_1 (2 + x)^7 (1 - x) + {}^8C_2 (2 + x)^6 (1 - x)^2 - \ldots\ldots\ldots + (1 - x)^8$

according to the ascending exponents of x after reducing the expansion.

Solution

The expression is forming the expansion of :

$\left((2 + x) - (1 - x)\right)^8 = (2 + x - 1 + x)^8 = (1 + 2x)^8$

$\therefore T_5 = {}^8C_4 (2x)^4 = 70 \times 16 x^4 = 1120 x^4$

Example 9

In the expansion of : $(1 + m x)^9$ according to the ascending exponents of x , if the coefficient of the third term is 144 and the value of the seventh term is 84 , find the value of each of : m and x where m is a positive integer.

Solution

$\because T_3 = {}^9C_2 (mx)^2 = 36 m^2 x^2$ $\therefore$ The coefficient of $T_3 = 36 m^2$

$\therefore 36 m^2 = 144$ $\therefore m^2 = 4$ $\therefore m = \pm 2$

, $\because$ m is a positive integer $\therefore m = 2$

, $\because T_7 = 84$ $\therefore {}^9C_6 (m x)^6 = 84$ $\therefore 84 (2x)^6 = 84$

$\therefore (2x)^6 = 1$ $\therefore 2x = \pm 1$ $\therefore x = \pm \frac{1}{2}$

Rule

1. $(x + a)^n + (x - a)^n = 2 (T_1 + T_3 + T_5 + \cdots\cdots)$

i.e. Double the summation of the terms of odd order in the expansion of $(x + a)^n$

The number of terms of the expansion $= \frac{n}{2} + 1$ if n is even

The number of terms of the expansion $= \frac{n + 1}{2}$ if n is odd

2. $(x + a)^n - (x - a)^n = 2 (T_2 + T_4 + T_6 + \cdots\cdots)$

i.e. Double the summation of the terms of even order in the expansion of $(x + a)^n$

The number of terms of the expansion $= \frac{n}{2}$ if n is even

The number of terms of the expansion $= \frac{n + 1}{2}$ if n is odd

Example 10

Simplify the expression : $(1+\sqrt{b})^4+(1-\sqrt{b})^4$, then find the value of the result when : $b=\sqrt{2}$

Solution

$$(1+\sqrt{b})^4+(1-\sqrt{b})^4 = 2\,(T_1+T_3+T_5) = 2\,(1+{}^4C_2\,(\sqrt{b})^2+{}^4C_4\,(\sqrt{b})^4)$$

$$= 2\,(1+6\,b+b^2) = 2+12\,b+2\,b^2$$

When : $b=\sqrt{2}$

$\therefore$ The result $= 2+12\sqrt{2}+2\,(\sqrt{2})^2 = 2+12\sqrt{2}+4 = 6+12\sqrt{2}$

The middle term in the expansion of : $(x+a)^n$

In the expansion of : $(x+a)^n$, the number of terms = $(n+1)$ terms.

1. If n is even :

Then the number of terms in the expansion is $(n+1)$ which is odd. Thus the expansion will contain only one middle term whose order is $\frac{n+2}{2}$

i.e. The middle term is $\left(T_{\frac{n}{2}+1}\right)$

2. If n is odd :

Then the number of terms in the expansion is $(n+1)$ which is even. Thus the expansion will contain two middle terms , whose orders are $\frac{n+1}{2}$ and $\frac{n+3}{2}$

i.e. The two middle terms are $\left(T_{\frac{n+1}{2}}\right.$ and the next one$\left.\right)$

Example 11

Find the middle term (or two terms) in the expansion of each of the following expressions :

1. $\left(2x^2-\frac{1}{x}\right)^{12}$ **2.** $(2a-1)^9$

Solution

1. The power $n = 12$ (even number)

$\therefore$ The expansion has only one middle term its order is $\frac{12+2}{2}=7$

$\therefore$ The middle term $= T_7 = T_{6+1} = {}^{12}C_6\left(-\frac{1}{x}\right)^6(2x^2)^6 = 924\times\frac{1}{x^6}\times 64\,x^{12} = 59136\,x^6$

2. The power $n = 9$ (odd number)

$\therefore$ The expansion has two middle terms and theirs orders are $\frac{9+1}{2}$, $\frac{9+3}{2}$ **i.e.** 5 , 6

$\therefore$ The first middle term $= T_5 = T_{4+1} = {}^9C_4\,(-1)^4\,(2a)^5 = 126\times 32\,a^5 = 4032\,a^5$

, the second middle term $= T_6 = T_{5+1} = {}^9C_5\,(-1)^5\,(2a)^4$

$$= 126\times(-1)\times 16\,a^4 = -2016\,a^4$$

Example 12

If the middle term in the expansion of : $\left(3x^2 + \frac{2}{3}x^{-1}\right)^8$ equals 17920 , what is the value of : x ?

Solution

$\because$ The power = 8 (even number)

$\therefore$ The expansion has only one middle term = $\frac{8+2}{2} = 5$

$\therefore$ The middle term = $T_5 = T_{4+1} = {}^8C_4 \left(\frac{2}{3}x^{-1}\right)^4 (3x^2)^4$

$= {}^8C_4 \times \frac{2^4}{3^4} x^{-4} \times 3^4 x^8 = {}^8C_4 \times 2^4 x^4$

$\therefore {}^8C_4 \times 16x^4 = 17920$ $\quad\quad$ $\therefore 70 \times 16x^4 = 17920$

$\therefore x^4 = 16$ $\quad\quad$ $\therefore x = \pm 2$

Example 13

Find the middle term in the expansion of : $(1+3x)^{10} - (1-3x)^{10}$

Solution

$\because (1+3x)^{10} - (1-3x)^{10} = 2(T_2 + T_4 + T_6 + T_8 + T_{10})$

$\therefore$ The middle term in the expansion = $2T_6 = 2\,{}^{10}C_5 (3x)^5 = 2 \times 252 \times 3^5 \times x^5 = 122472x^5$

Example 14

If the two middle terms in the expansion of : $(ax + b)^{2n+1}$ are equal where $n \in \mathbb{Z}^+$

Prove that : $x = \frac{b}{a}$

Solution

The orders of the two middle terms = $\frac{2n+1+1}{2}$, $\frac{2n+1+3}{2}$ i.e. $n+1$, $n+2$

, $\because$ the two middle terms are equal $\quad\quad$ $\therefore T_{n+1} = T_{n+2}$

$\therefore {}^{2n+1}C_n\, b^n a^{n+1} x^{n+1} = {}^{2n+1}C_{n+1}\, b^{n+1} a^n x^n$

, $\because {}^{2n+1}C_n = {}^{2n+1}C_{n+1}$ $\quad\quad$ $\therefore b^n a^{n+1} x^{n+1} = b^{n+1} a^n x^n$

, dividing the two sides by $b^n a^n x^n$ $\quad\quad$ $\therefore ax = b$ $\quad\quad$ $\therefore x = \frac{b}{a}$

Remark

If the two middle terms in the expansion $(x+y)^n$ are equal , then $x = y$

In example (14) $\because$ The two middle terms are equal $\quad\quad$ $\therefore ax = b$ $\quad\quad$ $\therefore x = \frac{b}{a}$